Finite-Time Swarm Tracking: Numerical Results

All quantities are normalized unless unit dimensions are specified. This report documents the rotating and compressing / expanding multi-agent star formation tracking scenarios with transient collision avoidance and connectivity preservation:

- **Experiment 1:** Multi-formation reconfiguring demonstration across four distinct geometric templates along a straight-line reference trajectory with Hungarian assignment.
- **Experiment 2:** Finite-time convergence, disagreement decay, and coupling gain evaluation under a rotating-compressing star formation ($\kappa = 2.0$).
- **Experiment 3:** Transient collision avoidance under rapid star compression ($\kappa = 12.0$, $a_0 = 1.70 \to a_\infty = 0.72$) comparing certified safe ($m = 60.0$) versus uncertified collision ($m = 1.0$).
- **Experiment 4:** Dual transient connectivity preservation under rapid star expansion ($\kappa = 12.0$, $a_0 = 1.70 \to a_\infty = 3.02$) comparing certified connected ($m = 60.0$) versus uncertified link breach ($m = 1.0$).

1 Model and parameters

The physical swarm consists of $n = 10$ agents in $\mathbb{R}^2$ governed by the closed-loop velocity command:

$$\dot{x}_i = u_i = \dot{q}_i - \delta_i - \mathrm{sig}^\beta(\delta_i) - \sum_{j \in \mathcal{N}_i(t)} w_{ij} M(\delta_i - \delta_j), \qquad \delta_i = x_i - q_i,$$

where $\mathrm{sig}^\beta(z) = \mathrm{sign}(z)|z|^\beta$, $M = mI_2$, $\beta = 0.5$, $d_{\text{safe}} = 1.0$ m, and communication radius $R_c = 6.0$ m. The state-dependent proximity weights are

$$w_{ij}(x,t) = \frac{a_{ij}(t)}{\|x_i - x_j\| + 0.05}, \qquad a_{ij}(t) = \begin{cases} 1 & \|x_i - x_j\| < R_c \ (i \neq j), \\ 0 & \text{otherwise}, \end{cases}$$

yielding the uniform lower bound $\underline{w} = 1/(R_c + 0.05) = 1/6.05 \approx 0.1653$ m^{-1}. Rather than requiring the entire formation diameter to lie inside a ball of radius R_c, global graph connectivity is maintained via a designated 15-edge backbone graph $\mathcal{H} = (\mathcal{V}, \mathcal{E}_\mathcal{H})$. For the 10-agent star template, $\mathcal{H}$ consists of the 5 inner pentagonal ring edges among the valley agents and 10 spoke edges connecting each star tip to its two adjacent inner valley neighbors:

$$\mathcal{E}_\mathcal{H} = \{(1,3),(3,5),(5,7),(7,9),(9,1),(0,1),(0,9),(2,1),(2,3),(4,3),(4,5),(6,5),(6,7),(8,7),(8,9)\}.$$

The maximum template backbone edge at unit scale is $\ell_{\mathcal{H},0} \approx 1.9497$ m, and the algebraic connectivity of the unweighted backbone Laplacian is $\lambda_2(L_\mathcal{H}) \approx 0.9316$.

The rotating and compressing target stations $q_i(t)$ follow:

$$r(t) = \begin{bmatrix} t \\ 2\sin(0.5t) \end{bmatrix}, \qquad \dot{r}(t) = \begin{bmatrix} 1 \\ \cos(0.5t) \end{bmatrix},$$
$$q_i(t) = r(t) + a(t)R(\omega t)p_i^0, \qquad \dot{q}_i(t) = \dot{r}(t) + \dot{a}(t)R(\omega t)p_i^0 + a(t)\dot{R}(\omega t)p_i^0,$$

where $p_i^0 = b_i[\cos\phi_i, \sin\phi_i]^T$, $\phi_i = \pi/2 + \pi i/5$, and $b_i = 2.8$ for even i and 1.22 for odd i ($i = 0, \ldots, 9$), centered such that $\sum_i p_i^0 = 0$. The baseline minimum pairwise distance is $DQ_0 \approx 1.4342$ m. Rotation operates at angular velocity $\omega = \pi/4$ rad/s (45°/s) via

$$R(\omega t) = \begin{bmatrix} \cos(\omega t) & -\sin(\omega t) \\ \sin(\omega t) & \cos(\omega t) \end{bmatrix}, \qquad \dot{R}(\omega t) = \omega \begin{bmatrix} -\sin(\omega t) & -\cos(\omega t) \\ \cos(\omega t) & -\sin(\omega t) \end{bmatrix}.$$

Formation scale contraction follows

$$a(t) = a_\infty + (a_0 - a_\infty)e^{-\kappa t}, \qquad a_0 = 1.70, \quad a_\infty = 0.72, \quad \dot{a}(t) = -\kappa(a_0 - a_\infty)e^{-\kappa t}.$$

For all $t \geq 0$, the station plan satisfies $d_q(t) \geq a_\infty DQ_0 \approx 1.0326$ m $> d_{\text{safe}} = 1.0$ m and the maximum backbone edge satisfies $\ell_{\mathcal{H}}(t) \leq a_0 \ell_{\mathcal{H},0} = 3.3144$ m $< R_c = 6.0$ m. Thus, target stations remain strictly collision-free and connected along the backbone graph for all time.

Initial agent states are generated by random annular ring perturbations around each assigned station $q_i(0)$ with radius $r_i \in [0.05, 0.25]$ m ($d_{\text{safe}} = 1.0$ m, $R_c = 6.0$ m, seed 123), with an inward push of 0.55 m applied along closest station pair $(1, 3)$, without enforcing zero-mean centering ($\sum_i \delta_i(0) \neq 0$). The disagreement vector is formed internally via $\eta_i(0) = \delta_i(0) - \frac{1}{n}\sum_j \delta_j(0)$. At $t = 0$, initial physical separation is safe ($d_x(0) > d_{\text{safe}}$) and connected ($\ell_x(0) < R_c$), with initial error norm $\|\delta_0\| \approx 0.9634$, disagreement distortion $\rho_0 = \sqrt{2}\|\eta_0\| \approx 1.3303$ m, and universal settling deadline $\overline{T} = 2\ln(1 + \sqrt{\|\delta_0\|}) \approx 1.3621$ s.

The geometric clearance margin along the backbone is $s(t) = \min\{d_q(t) - d_{\text{safe}},\ R_c - \ell_{\mathcal{H}}(t)\} = \min\{a(t)DQ_0 - d_{\text{safe}},\ R_c - a(t)\ell_{\mathcal{H},0}\}$. At $t = 0$, $s(0) = \min\{1.4381\text{ m}, 2.6856\text{ m}\} = 1.4381$ m, ensuring $\rho_0 < s(0)$ is strictly satisfied with positive clearance margin $s(0) - \rho_0 \approx 0.1078$ m. For coupling gain m, the effective decay rate is $\gamma = 1 + m\underline{w}\lambda_2(L_{\mathcal{H}}) = 1 + m\frac{0.9316}{6.05} \approx 1 + 0.1540m$, and the required rate is

$$\gamma_{\text{req}}(\kappa) = \sup_{\tau \in (0, \overline{T}]} \frac{\ln(\rho_0 / s(\tau))}{\tau}.$$

The dynamic disagreement distortion tube is bounded by

$$\rho_{\mathcal{H}}(t) = \min\{\sqrt{2}R(t),\ \rho_0 e^{-\gamma t}\}, \qquad R(t) = \max\left\{0, \left((1 + \sqrt{\|\delta_0\|})e^{-\frac{1}{2}t} - 1\right)^2\right\},$$

where $R(t)$ is the pure station tracking bound (independent of graph and coupling gain m).

2 Formation demonstration

The multi-formation reconfiguring demonstration (Exp. 1) tracks reconfiguring formations along a straight-line reference trajectory $r(t) = [v_x t, 0]^T$ ($v_x = 1.6$ m/s). Inter-station transitions across circle, wedge, star, and arrow templates are optimized using Hungarian assignment ($\min \sum \|q_i^+ - q_i^-\|^2$), reducing average slot displacement by over 65% and preventing unnecessary trajectory intertwining.

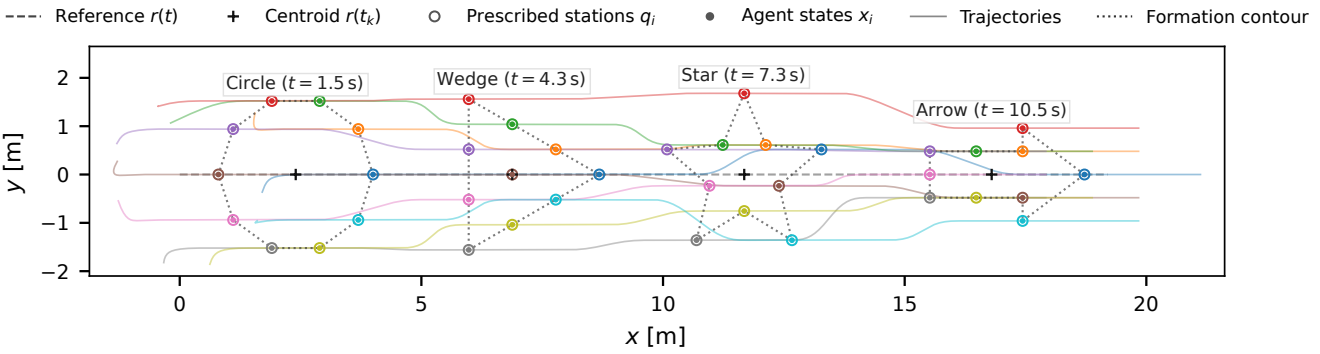


Figure 1: Experiment 1: Reconfiguring formations along a straight-line reference trajectory. Target formations (Circle at $t = 1.5$ s, Wedge at $t = 4.3$ s, Star at $t = 7.3$ s, Arrow at $t = 10.5$ s) are highlighted with dotted contours, assigned target stations (circles), agent positions (dots), and continuous tracking trajectories.

3 Finite-time tracking and coupling gain evaluation

Experiment 2 evaluates the rotating-compressing star formation maneuver ($\kappa = 2.0$) across coupling gains $m \in \{0, 0.5, 2, 5, 10\}$. Figure 2 depicts station tracking error $\|\delta(t)\|_2$, formation disagreement $\|\eta(t)\|_2$, and the initial control command $\|u(t)\|_2$. In panels (a) and (b), error curves are cut at 10^{-4} to illustrate finite-time settling.

Increasing the coupling gain m accelerates disagreement decay, reducing the settling time to the 10^{-6} error threshold from 1.1517 s at $m = 0$ down to 0.3933 s at $m = 10$ (Table 1). All runs settle well within the theoretical universal deadline $\overline{T} = 1.3621$ s.

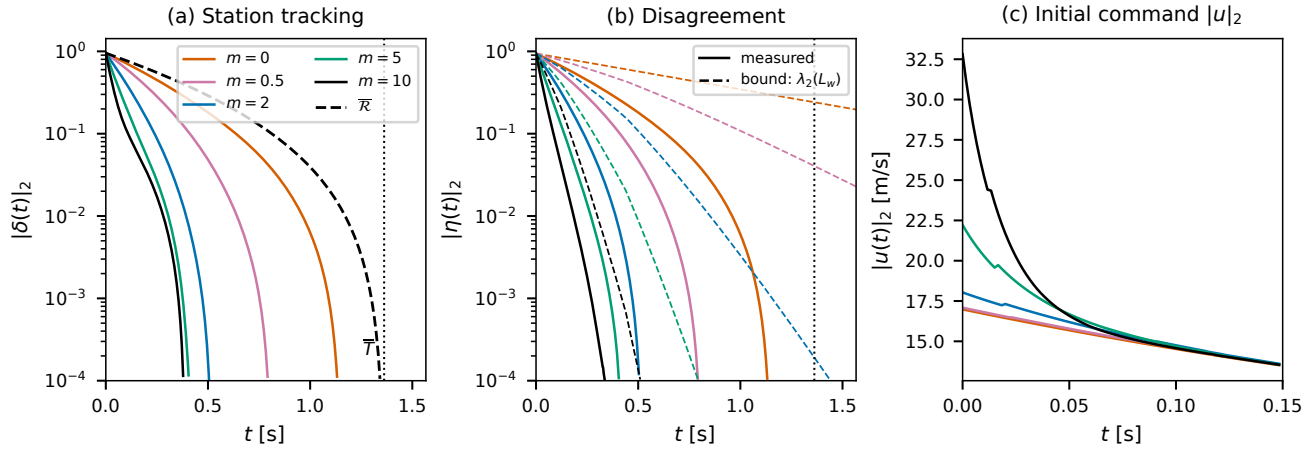


Figure 2: Experiment 2 finite-time tracking and disagreement across coupling gains m ($\kappa = 2.0$). (a) Station tracking $\|\delta\|_2$ against universal bound $R(t)$ and deadline $\overline{T}$, cut at 10^{-4}. (b) Disagreement $\|\eta\|_2$ against exponential rate bound, cut at 10^{-4}. (c) Initial control command $\|u\|_2$ ($t \in [0, 0.15]$ s).

Table 1: Experiment 2: Settling times and control metrics across coupling gains m under rotating star formation ($\kappa = 2.0$).

m	$t_{10^{-6}}$ [s]	$\overline{T}$ [s]	Peak $\|u\|_2$	J_u
0.0	1.1517	1.3621	16.9828	429.44
0.5	0.8117	1.3621	17.0860	429.92
2.0	0.5250	1.3621	18.0322	431.55
5.0	0.4467	1.3621	22.2186	434.95
10.0	0.3933	1.3621	32.8156	440.74

4 Star formation safety and connectivity transient

Experiments 3 and 4 analyze aggressive maneuvers sharing identical rate $\kappa = 12.0$ and gains ($m = 60.0$ for the safe regime, $m = 1.0$ for the unsafe regime), differentiated solely by the scale limits. Experiment 3 evaluates fast compression ($a_0 = 1.70 \to a_\infty = 0.72$) over $t \in [0, 2.5]$ s, comparing two regimes in a 2×3 grid format (Figure 3):

- **Row 1 (Certified Safe Case, $m = 60.0$):** With coupling gain $m = 60.0$, the convergence rate is $\gamma = 10.2388$, strictly exceeding the theoretical threshold $\gamma_{\text{req}} = 9.9879$ ($\Delta_{\min} = 0.0084$ m > 0). In closed-loop simulation the swarm remains strictly collision-free throughout the transient ($d_{x,\min} = 1.0326$ m $> d_{\text{safe}} = 1.0$ m) with backbone links preserved ($\ell_{x,\max} = 3.9437$ m $< R_c$), settling by $t_{\text{set}} = 0.1767$ s and certifying outcome `Certified; Safe, Connected`.
- **Row 2 (Unsafe Collision Case, $m = 1.0$):** With low coupling gain ($m = 1.0$), the decay rate is only $\gamma = 1.1540 \ll \gamma_{\text{req}}$. Rapid maneuver compression severely outpaces error convergence, resulting in an observed physical collision mid-transient: $d_x(t) < d_{\text{safe}}$ for $t \in [0.04, 0.30]$ s (duration 0.26 s), reaching $d_{x,\min} = 0.8923$ m (a collision penetration of 10.77 cm), yielding outcome `Uncertified; Safety Violation`.

Figure 3 presents the 2×3 transient response comparison. Column 1 displays the minimum pairwise distance $d_x(t)$ against the lower tube bound $d_q(t) - \rho_{\mathcal{H}}(t)$ and threshold $d_{\text{safe}} = 1.0$ m, with the collision breach shaded in red for Row 2. Column 2 shows maximum pairwise distance $\ell_x(t)$ against $\ell_q(t) + \rho_{\mathcal{H}}(t)$ and connectivity limit $R_c = 6.0$ m. Column 3 compares geometric margin $s(t)$ against distortion tube radius $\rho_{\mathcal{H}}(t)$, highlighting certified ($s \geq \rho_{\mathcal{H}}$) in green and uncertified ($s < \rho_{\mathcal{H}}$) in orange.

Table 2: Experiment 3 outcomes: Certified Safe ($m = 60.0$) vs. collision violation ($m = 1.0$) under rapid star compression ($a_0 = 1.70 \to a_\infty = 0.72$, $\kappa = 12.0$, $t \in [0, 2.5]$ s).

Case	m	κ	γ	γ_{req}	t_{set} [s]	$d_{x,\min}$	$\ell_{x,\max}$	Peak $\|u\|$	J_u	Outcome
Safe	60.0	12.0	10.2388	9.9879	0.1767	1.0326	3.9437	178.6172	678.85	Certified; Safe, Connected
Unsafe	1.0	12.0	1.1540	9.9879	0.5450	0.8923	3.9437	80.6047	626.34	Uncertified; Safety Violation

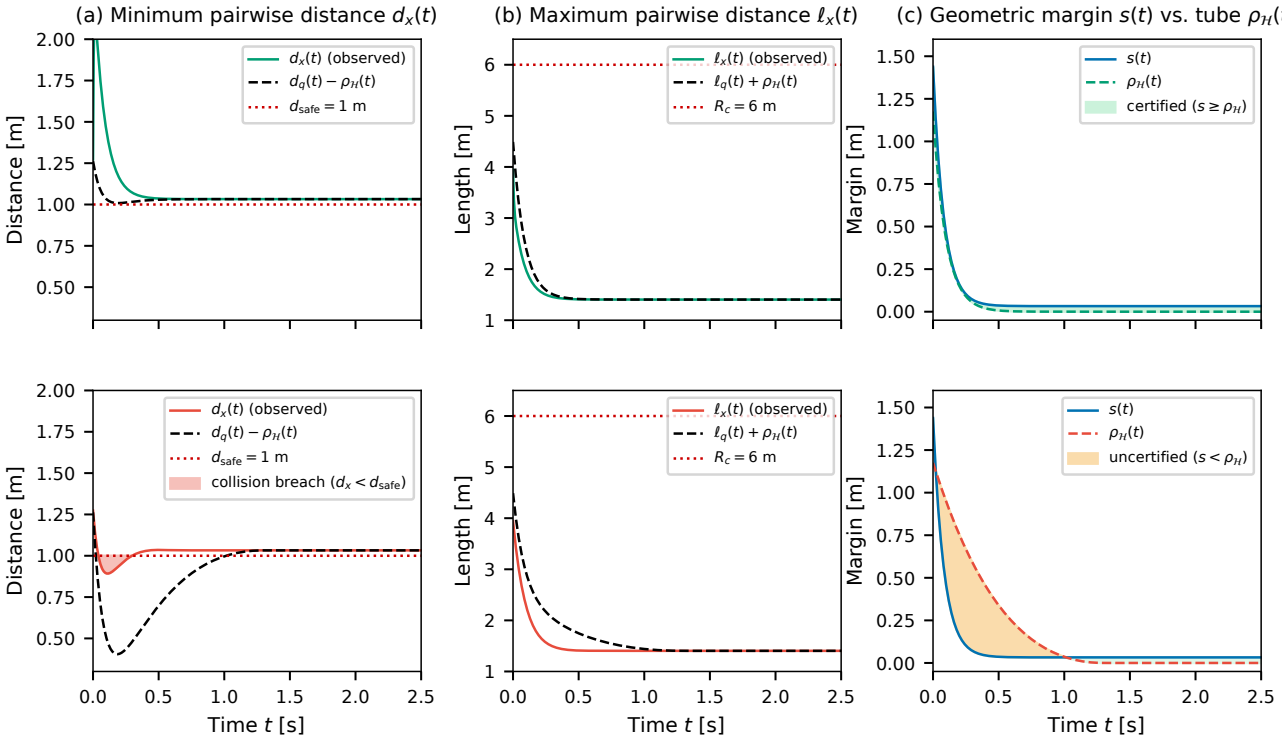


Figure 3: Experiment 3: Safety and connectivity during fast star compression ($a_0 = 1.70 \to a_\infty = 0.72$, $\kappa = 12.0$, $t \in [0, 2.5]$ s). Top row: Certified Safe ($m = 60.0$). Bottom row: Unsafe Collision Violation ($m = 1.0$). Columns show minimum pairwise distance $d_x(t)$ (left), maximum pairwise distance $\ell_x(t)$ (middle), and geometric margin $s(t)$ vs. distortion tube $\rho_{\mathcal{H}}(t)$ (right).

5 Dual expansion maneuver and transient connectivity preservation

Complementary to rapid compression (Exp. 3), Experiment 4 tests a rapid expansion maneuver ($a_0 = 1.70 \to a_\infty = 3.02$ at $\kappa = 12.0$) to evaluate network connectivity preservation ($\ell_x(t) \leq R_c = 6.0$ m). As the star expands, nominal backbone edges stretch to $a_\infty \ell_{\mathcal{H},0} = 5.888$ m, leaving a narrow margin $s_{\text{conn}}(\infty) = 0.112$ m. The required rate is $\gamma_{\text{req}} = 6.3408$.

Table 3: Experiment 4 outcomes: Certified Safe & Connected ($m = 60.0$) vs. uncertified disconnection ($m = 1.0$) under rapid star expansion ($a_0 = 1.70 \to a_\infty = 3.02$, $\kappa = 12.0$, $t \in [0, 2.5]$ s).

Case	m	κ	γ	γ_{req}	t_{set} [s]	$d_{x,\min}$	$\ell_{x,\max}$	Peak $\|u\|$	J_u	Outcome
Safe	60.0	12.0	10.2388	6.3408	0.2417	1.8351	5.8880	148.1963	3803.88	Certified; Safe, Connected
Unsafe	1.0	12.0	1.1540	6.3408	1.0433	1.8351	6.2417	108.2098	3790.96	Uncertified; Connectivity Violation

Figure 4 presents the corresponding 2×3 grid:

- **Row 1 (Certified Safe & Connected, $m = 60.0$):** With coupling gain $m = 60.0$, $\gamma = 10.2388 > \gamma_{\text{req}} = 6.3408$. The certificate is satisfied ($\Delta_{\min} = 0.1113$ m > 0). All backbone communication links remain strictly within range ($\ell_{x,\max} = 5.8880$ m $< R_c$), settling by $t_{\text{set}} = 0.2417$ s with outcome `Certified; Safe, Connected`.

- **Row 2 (Unsafe Transient Disconnection, $m = 1.0$):** With low coupling gain ($m = 1.0$, $\gamma = 1.1540 < \gamma_{\text{req}}$), tracking lag under rapid expansion causes backbone edge stretch exceeding range: $\ell_x(t) > R_c = 6.0$ m during $t \in [0.12, 0.58]$ s (duration 0.46 s, peak $\ell_{x,\max} = 6.2417$ m), suffering a genuine connectivity violation (`Uncertified; Connectivity Violation`).

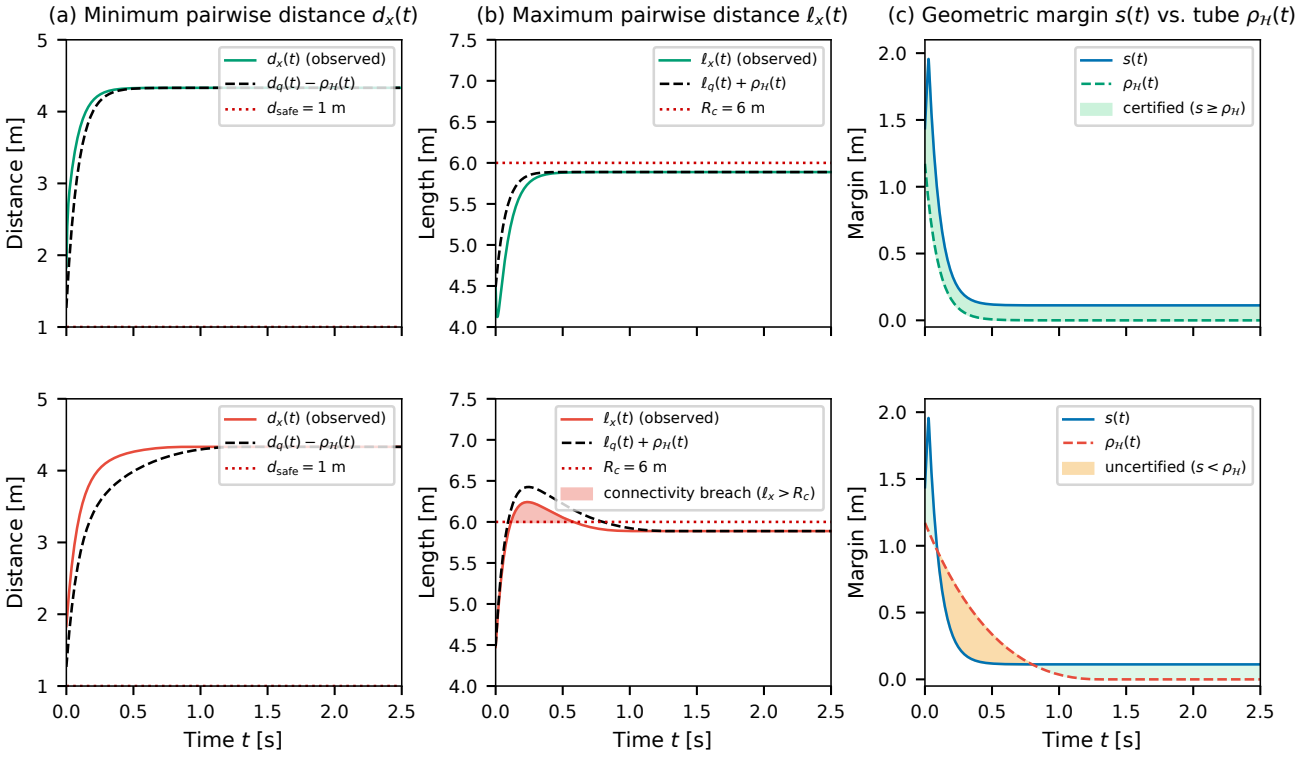


Figure 4: Experiment 4: Safety and connectivity during fast star expansion ($a_0 = 1.70 \to a_\infty = 3.02$, $\kappa = 12.0$, $t \in [0, 2.5]$ s). Top row: Certified Safe & Connected ($m = 60.0$). Bottom row: Unsafe Transient Disconnection ($m = 1.0$). Columns show minimum pairwise distance $d_x(t)$ (left), maximum pairwise distance $\ell_x(t)$ (middle), and geometric margin $s(t)$ vs. distortion tube $\rho_{\mathcal{H}}(t)$ (right).

6 Reproduction

All experiments in this document can be reproduced using:

- Experiment 1: `python3 run_experiments.py --experiment 1`
- Experiment 2: `python3 run_experiments.py --experiment 2`
- Experiment 3: `python3 run_experiments.py --experiment 3`
- Experiment 4: `python3 run_experiments.py --experiment 4`

Compile the document from `ECC_paper/` using `pdflatex numerical_results.tex`.